Econometrics I

Lecture 8

Heteroscedasticity-robust estimation

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Heteroskedasticity

Heteroskedasticity is the word denoting that assumption $\text{var}(u_i|X) = \mathbb{E}(u_i^2|X) = \sigma^2$ does not hold.

Learning goals:

- Recall: How to detect heteroskedasticity?
  - Visual inspection
  - Residual diagnostic with F-test
- What to do against it?
  - Data transformation
  - Heteroskedasticity-robust standard errors
  - Alternative to OLS: weighted or generalized least squares

Recall...

Variance of OLS Estimator

Recall that the classical linear regression model assumes:

$$
\begin{aligned}
\mathrm{E}[uu'|X] &= \begin{bmatrix} \mathrm{E}[u_1^2|X] & \mathrm{E}[u_1u_2|X] & \cdots & \mathrm{E}[u_1u_n|X] \\ \mathrm{E}[u_2u_1|X] & \mathrm{E}[u_2^2|X] & \cdots & \mathrm{E}[u_2u_n|X] \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{E}[u_nu_1|X] & \mathrm{E}[u_nu_2|X] & \cdots & \mathrm{E}[u_n^2|X] \end{bmatrix} \\
&= \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 I_{n\times n}
\end{aligned}
$$

Where:

- **No autocorrelation**: Off-diagonal elements of (1) equal zero
- **Homoscedasticity**: Diagonal elements of (1) equal σ^2

Variance of OLS Estimator

Combining with correct specification (S2) and exogenous zero mean error (S3), the variance of $\hat{\beta} = (X'X)^{-1}X'y$ is:

$$\begin{aligned}
\mathbb{V}(\hat{\beta}|X) &= \mathbb{E}\left[(\hat{\beta} - \mathbb{E}[\beta])(\hat{\beta} - \mathbb{E}[\beta])'|X\right] \\
&= \mathbb{E}\left[((X'X)^{-1}X'u)((X'X)^{-1}X'u)'|X\right] \\
&= \mathbb{E}\left[((X'X)^{-1}X'u)(u'X(X'X)^{-1})|X\right] \\
&= \mathbb{E}\left[(X'X)^{-1}X'uu'X(X'X)^{-1}|X\right] \\
&= (X'X)^{-1}X'\mathbb{E}(uu'|X)X(X'X)^{-1}
\end{aligned}$$

Variance of OLS Estimator

With assumption S4, $\mathbb{E}(uu') = \sigma^2 I$, then:

$$\begin{aligned} \mathbb{V}(\hat{\beta}_{\textbf{OLS}}|X) &= (X'X)^{-1}X'\mathbb{E}(uu')X(X'X)^{-1} \\ &= (X'X)^{-1}X'\sigma^2 I_{n\times n}X(X'X)^{-1} \\ &= \sigma^2(X'X)^{-1} \end{aligned} \quad (1)$$

The resulting $k \times k$ matrix in (1) contains the k variances of the k regressors in vector $\hat{\beta}$.

Consequences of Violation

Assumption	Means	Required for
1: $\text{rank}(X) = k$	No perfect multicollinearity	Existence of $\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y$
2: $y = X\beta + u$	Correct functional specification	Unbiasedness: $\mathbb{E}[\hat{\beta}\|X] = \beta$ (also 1 & 3 needed)
3: $\mathbb{E}[u\|X] = 0$	Exogenous zero-mean errors	Unbiasedness: $\mathbb{E}[\hat{\beta}\|X] = \beta$ (also 1 & 2 needed)
4: $\mathbb{V}[u\|X] = \sigma^2 I_n$	Homoskedastic & nonautocorrelated errors	Formula $\mathbb{V}(\hat{\beta}\|X) = \sigma^2(X'X)^{-1}$ (also 1, 2 & 3 needed)
5: $u\|X \sim N(0, \sigma^2 I_n)$	Normally-distributed errors	Validity of t-tests, confidence intervals, etc. (also 1 - 4 needed)

Consequences of Violation

What happens if $\mathbb{E}(uu'|X) \neq \sigma^2 I$? Naturally, the variance (1) will be poorly estimated and significances will be invalid.

Suppose $\mathbb{E}(uu'|X) = \Omega$, where Ω is a matrix with different values both on and off the diagonal (representing heteroscedastic and autocorrelated errors).
In this case, the variance-covariance matrix is not given by (1) but by:

$$\begin{aligned} \mathbb{V}(\hat{\beta}|X) &= (X'X)^{-1} X' \mathbb{E}(uu'|X) X (X'X)^{-1} \\ &= (X'X)^{-1} X' \Omega X (X'X)^{-1} \qquad (2) \end{aligned}$$

Sometimes sufficient: Variable transformations

Heteroscedasticity example

Figure: Food expenditure vs. income (by household)

Figure: OLS Estimation

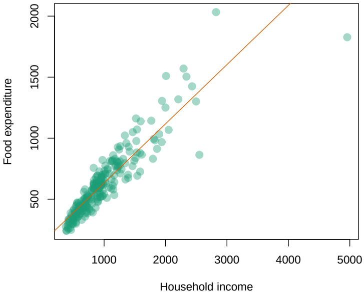
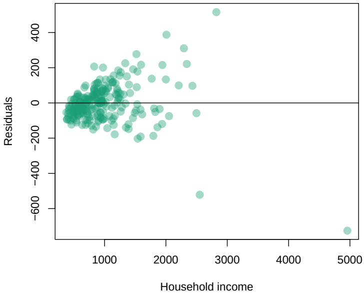


Figure: Estimation Residuals



The shown relationship is clearly heteroscedastic.

Log Transformation

Instead of applying complex methods, it's advisable to start with simple variable transformations. In our example, if we apply the logarithm to both **income** and **foodexp**, we obtain a relationship with linearity and absence of heteroscedasticity—at least apparently.

Log Transformation

Figure: Log-log Estimation

Figure: OLS Fit

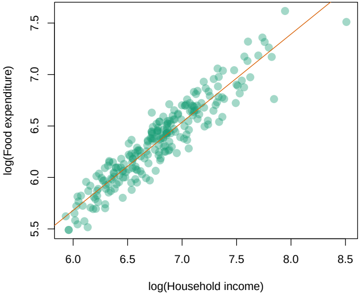
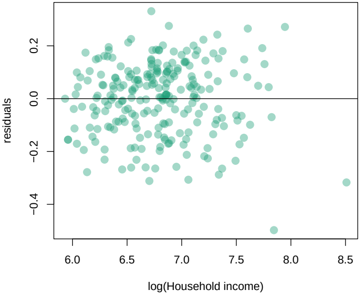


Figure: Residuals



Sometimes the most convenient way to combat heteroscedasticity is simply to transform using ln(z), exp(z), z^2, or $1/z$, among others, which can alleviate heteroscedasticity present in the linear model.

Log Transformation

But are residuals truly homoskedastic. Let's test with an order 3 polynomial.

```
OLS <- lm(foodexp ~ income, data=engel) # Level-level OLS regression (name =
    OLS)

Log <- lm(log(foodexp) ~ log(income), data=engel) # Log-log OLS regression (
    name = Log)

homoTest_OLS <- lm(residuals(OLS)^2 ~ income  + I(income^2)  + I(income^3)
    , data=engel)
homoTest_Log <- lm(residuals(Log)^2 ~ log(income) + I(log(income)^2)+ I(log(
    income)^3), data = engel)

stargazer(homoTest_OLS,homoTest_Log,type="text")
```

Log Transformation

	Dependent variable:	
	residuals(OLS)2 (1)	residuals(Log)2 (2)
regressor	−61.256***	1.458
	(18.933)	(1.119)
regressor^2	0.040***	−0.226
	(0.011)	(0.159)
regressor^3	−0.00000	0.012
	(0.00000)	(0.008)
Constant	27,004.790***	−3.126
	(9,060.716)	(2.608)
Observations	235	235
R2	0.771	0.117
Adjusted R2	0.769	0.105
Residual Std. Error (df = 231)	21,237.130	0.025
F Statistic (df = 3; 231)	259.962***	10.181***
Note:		*p<0.1; **p<0.05; ***p<0.01

(Your output should look slightly different but with the same results)

Log Transformation

- Model 2 much less heteroscadastic than model 1 (compare F-statistic). Yet ...
- H_0 : "homoscedastic errors" was rejected even using the log-log model (significant F-statistic).
- What can be done?
- The simplest solution: substitute st. errors by so-called heteroscedasticity-robust standard errors.

Alternative 1: OLS with Robust Standard Errors

Robust Variance Estimation

We've seen that with heteroscedasticity, the OLS estimator:

$$\hat{\beta}_{\textbf{OLS}} = (X'X)^{-1}X'y \tag{3}$$

remains unbiased but the variance-covariance matrix of $\hat{\beta}_{\textbf{OLS}}$ is:

$$\mathbb{V}(\hat{\beta}|X) = (X'X)^{-1}X'\Omega X(X'X)^{-1} \tag{4}$$

Robust Variance Estimation

Only in the particular case of $\Omega = \sigma^2 I$ is the classical calculation valid via:

$$\mathbb{V}(\hat{\beta}|X) = \sigma^2(X'X)^{-1} \tag{5}$$

In the presence of heteroscedasticity, a simple alternative widely used in practice is the estimation of $\hat{\beta}$ via (3) but using a **heteroscedasticity-robust** variance-covariance matrix $\widehat{\Omega}$ substituting for Ω in (4).

White's Correction

There are multiple ways to obtain $\widehat{\Omega}$, with the most common being White's correction, where $\widehat{\Omega}$ is a diagonal matrix with elements $\hat{\sigma}_i^2$ that correspond simply to the squared residuals obtained from a simple OLS regression according to (3).

White standard errors are obtained in two steps:

1. Regress y on X to obtain squared residuals $\hat{u}_i^2$
2. Impute $\hat{u}_i^2$ into the diagonal of $\widehat{\Omega}$ and extract standard errors from matrix (4).

White's Correction

```r
# R code for robust standard errors
library(sandwich)
library(lmtest)

# OLS regression
model <- lm(y ~ x1 + x2, data = df)

# White robust standard errors
white_se <- sqrt(diag(vcovHC(model, type = "HC0")))

# Display results with robust SEs
coeftest(model, vcov = vcovHC(model, type = "HC0"))
```

Common HCCME versions:

- HC_0: $\hat{u}_i^2$ (White standard errors)
- HC_1: $\hat{u}_i^2 \cdot n/(n-k)$
- HC_2: $\hat{u}_i^2/(1-h_i)$ where h_i = leverage (x-ouliers)
- HC_3: $\hat{u}_i^2/(1-h_i)^2$

White's Correction

	Dependent variable: Food expenditure		
	OLS (classic)	OLS (White se)	OLS (HC3 se)
income	0.485***	0.485***	0.485***
	(0.014)	(0.052)	(0.066)
Constant	147.475***	147.475***	147.475**
	(15.957)	(46.449)	(59.955)
Observations	235	235	235
R2	0.830	0.830	0.830
Note:	*p<0.1; **p<0.05; ***p<0.01		

Same coefficients but different st. errors $\implies$ different t-statistics.

Leverage

Recall HCCME versions:

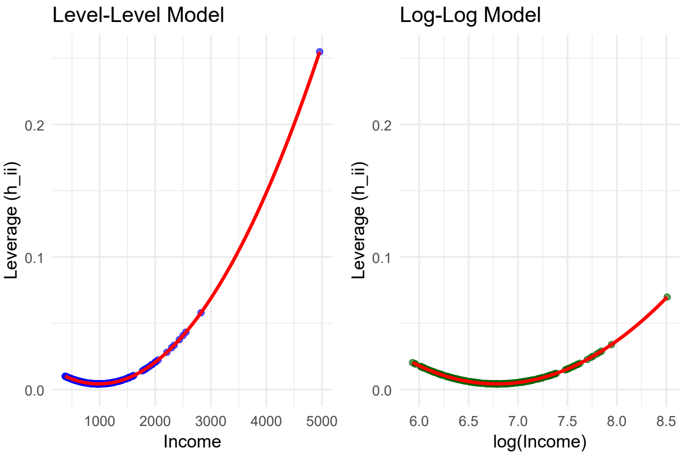
- HC_0: $\hat{u}_i^2$ (White standard errors)
- HC_1: $\hat{u}_i^2 \cdot n/(n-k)$
- HC_2: $\hat{u}_i^2/(1-h_i)$ where h_i = leverage (x-ouliers)
- HC_3: $\hat{u}_i^2/(1-h_i)^2$

What is Leverage?

- Leverage h_i measures how far an observation's predictors are from the mean of all predictors
- Diagonal elements of the 'hat matrix' $H = X(X'X)^{-1}X'$
- Range: $0 \leq h_i \leq 1$
- Rule of thumb: $h_i > \frac{2k}{n}$ indicates high leverage

Leverage

Figure: Leverage (h_i) vs independent variable

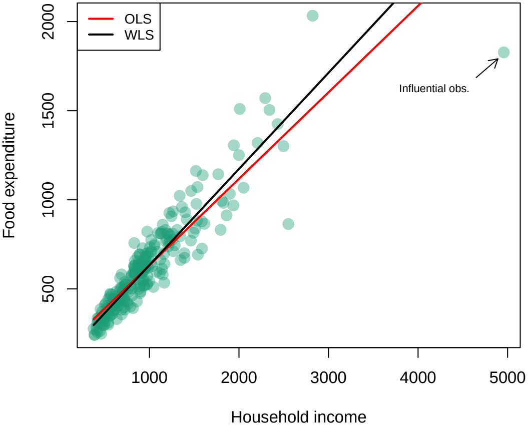


- Log model less affected by high leverage.

Diagnosis
Heteroskedasticity
Recall...
Variance of OLS Estimator
Consequences of Violation
Sometimes sufficient: Variable trans-formations
Heteroscedasticity example
Log Transformation
Alternative 1: OLS with Robust Standard Errors
Robust Variance Estimation
White's Correction
Leverage
Alternative 2: Feasible Generalized

Leverage

Figure: Leverage effect on OLS estimate



OLS highly affected by high-leverage point. Now we will see an alternative estimator (WLS) that is more robust to such points.

Alternative 2: Feasible Generalized Least Squares

The Ideal Case: Known Omega

The ideal case of $\mathbb{E}[uu'|X] \neq \sigma^2 I$ is one where the data generating process is:

$$y = X\beta + u, \qquad \mathbb{E}[uu'|X] = \Omega, \tag{6}$$

and the error variance-covariance matrix Ω **is known**. In this case, since Ω is a symmetric matrix, it's always possible to find Ψ such that $\Omega^{-1} = \Psi\Psi'$.

The Ideal Case: Known Omega

We can multiply y by Ψ':

$$\begin{aligned} \Psi' y &= \Psi' X \beta + \Psi' u \\ y^\star &= X^\star \beta + u^\star \end{aligned} \tag{7}$$

and obtain:

$$\begin{aligned} \hat{\beta}_{\textbf{GLS}} &= (X^{\star\prime} X^\star)^{-1} X^{\star\prime} y^\star \\ &= (X' \Psi \Psi' X)^{-1} X' \Psi \Psi' y = (X' \Omega^{-1} X)^{-1} X \Omega^{-1} y \end{aligned} \tag{8}$$

The Ideal Case: Known Omega

Generalized Least Squares (GLS) Estimator:

$$\hat{\beta}_{\textbf{GLS}} = (X'\Omega^{-1}X)^{-1}X\Omega^{-1}y \tag{9}$$

The particular aspect of this estimator is that the errors $u^\star = \Psi' u$ are characterized by:

$$\begin{aligned}\mathbb{E}[u^\star u^{\star\prime}|X] = \mathbb{E}[\Psi' u(\Psi' u)'|X] &= \mathbb{E}[\Psi' uu'\Psi|X] \\ &= \Psi'\mathbb{E}[uu'|X]\Psi = \Psi'\Omega\Psi = I\end{aligned}$$

Proof of last equality:

$$\Psi'\Omega\Psi = I \Leftrightarrow \Psi\Psi'\Omega\Psi = \Psi \Leftrightarrow \Omega^{-1}\Omega\Psi = \Psi$$

The Ideal Case: Known Omega

Therefore:

$$\mathbb{V}[\hat{\beta}_{\textbf{GLS}}|X] = (X^{\star\prime}X^{\star})^{-1} = (X'\Psi\Psi'X)^{-1} = (X'\Omega^{-1}X)^{-1} \tag{10}$$

Note:

- The j-th element of the diagonal of (10) corresponds to the variance of coefficient $\hat{\beta}_j$ and, maintaining the normality assumption, significance can be judged from the ratio $t = \hat{\beta}_j/\sqrt{\mathrm{var}[\hat{\beta}_j]}$ analogous to CLRM.
- Ω allows to incorporate both heteroscedasticity (today's class) and autocorrelation (more later on).

FGLS example: WLS estimator

In practice, we use the **Feasible Generalized Least Squares (FGLS)** method, which differs from GLS by stating that the matrix Ω of (2) **is not known but can be estimated**.

Assume all CLRM assumptions hold except homoscedasticity. In this case, the error variance-covariance matrix will have off-diagonal elements equal to zero (since no autocorrelation still holds) and diagonal elements $\sigma_i^2 \neq \sigma^2$ different for each i.

FGLS example: WLS estimator

That is:

$$\mathbb{E}[uu'|X] = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{bmatrix} \tag{11}$$

FGLS example: WLS estimator

Suppose, for example, that the DGP is given by:

$$y_i = x_{(i)}\beta + u_i, \qquad \mathbb{E}[u_i^2] = \exp(Z_i\gamma), \tag{12}$$

for each observation i. That is, the variance of u is a function of factors included in matrix Z [$\exp(\cdot)$ ensures that $\widehat{\hat{u}_i^2} > 0$], which may or may not partially coincide with X.
One could model the variance of u with:

$$\ln \hat{u}_i^2 = Z_{(i)}\gamma + \nu_i,$$

and impute each estimated value $\hat{u}_i^2$ into $\widehat{\Omega}$.

FGLS example: WLS estimator

Then, the estimator for β would be analogous to (9):
Feasible Generalized Least Squares (FGLS) Estimator:

$$\hat{\beta}_{\textbf{FGLS}} = (X'\widehat{\Omega}^{-1}X)^{-1}X\widehat{\Omega}^{-1}y \tag{13}$$

Its variance-covariance matrix will simply be:

$$\widehat{\mathbb{V}}[\hat{\beta}_{\textbf{FGLS}}|X] = (X'\widehat{\Omega}^{-1}X)^{-1}. \tag{14}$$

FGLS example: WLS estimator

If we know that $\widehat{\Omega}$ has zero off-diagonal elements, then its inverse corresponds to:

$$\widehat{\Omega}^{-1} = \begin{bmatrix} 1/\hat{\sigma}_1^2 & 0 & \cdots & 0 \\ 0 & 1/\hat{\sigma}_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\hat{\sigma}_n^2 \end{bmatrix} \tag{15}$$

FGLS example: WLS estimator

In this case, observation i of the transformation presented in (7) will be:

$$\begin{aligned} y_i^\star &= x_{(i)}^\star \beta + u_i^\star \\ y_i/\hat{\sigma}_i &= (1/\hat{\sigma}_i) x_{(i)} \beta + u_i/\hat{\sigma}_i \end{aligned} \tag{16}$$

The OLS estimator of regression (16) is called the **Weighted Least Squares (WLS) estimator**, since $w_i = 1/\hat{\sigma}_i$ functions as a weight for the regression.

WLS Application Example

```r
# R code for Weighted Least Squares
# Step 1: Estimate variance function
ols_model <- lm(foodexp ~ income, data = engel)
engel$u <- residuals(ols_model)
engel$u2 <- engel$u^2

# Estimate variance model (no constant to ensure >
    0)
var_model <- lm(u2 ~ 0 + income, data = engel)
engel$sigma2_hat <- fitted(var_model)

# Step 2: WLS regression
wls_model <- lm(foodexp ~ income, data = engel,
                weights = 1/sigma2_hat)
```

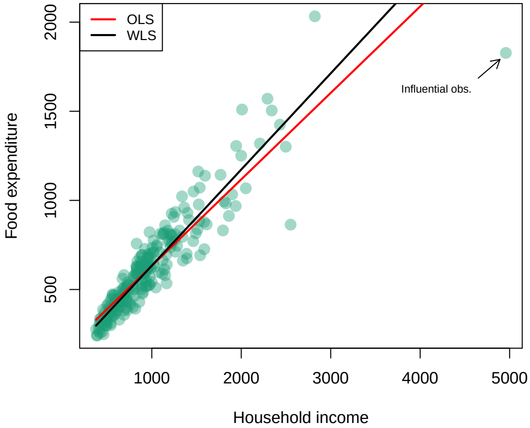
WLS Application Example

Dependent variable:	Food expenditure			ys
	OLS	OLS (White SE)	WLS 1	WLS 2
income	0.485***	0.485***	0.540***	
	(0.014)	(0.052)	(0.014)	
xs				0.540***
				(0.014)
cs				94.095***
				(12.917)
Constant	147.475***	147.475***	94.095***	
	(15.957)	(46.648)	(12.917)	
Observations	235	235	235	235
R2	0.830	0.830	0.856	0.979
Note:		*p<0.1;	**p<0.05;	***p<0.01

- The two WLS procedures produce equivalent results.
- Note higher slope than with OLS. Why? ...

WLS Application Example

Figure: FGLS Application with WLS



Observations with high $\hat{\sigma}_i$ (high **income** in our example) will have low value of $w_i = 1/\hat{\sigma}_i$, weighting less in the regression.
The figure's marked observation generates less leverage when estimated with weighted least squares.

TAKEAWAYS

Key takeaways

Studied approaches to deal with $\mathbb{V}(u|X) \neq \sigma^2 I_n$:

1. **Variable transformations**: Try log, square root, or other monotonic transformations to stabilize variance (or, alternatively, to reduce h_i).
2. **Robust standard errors**: Maintain OLS coefficients but use White/HC corrections of s.e.
3. **GLS**: ideal theoretical case, when Ω known. Allows incorporating both heteroscedasticity (today's class) and autocorrelation (more later on).
4. **Feasible GLS**: estimated $\widehat{\Omega}$ instead. Application: **Weighted Least Squares**:
   - variance function estimated from the data.
   - downweight high-variance observations